Amateur Baseball Pitcher Profiler: Decomposing Playing Style via Non-Negative Matrix Factorization

Octavian Yuen

University of California, Berkeley

Advised by Professor Bryan Graham

Department of Economics, UC Berkeley

Accepted for Poster Research Presentation for the 2026 SABR Analytics Conference

Abstract

Pitcher evaluation in amateur baseball presents a unique analytical challenge. Seasons are short, scouting resources are limited, and decisions often must be made from sparse and imperfect tracking data. Traditional clustering methods, while effective at the professional level, force pitchers into discrete categories and produce latent components that can be difficult to communicate to coaches and scouts without statistical training. We propose Non-Negative Matrix Factorization (NMF) as a principled alternative for pitcher archetype discovery in amateur settings. By decomposing a non-negative matrix of pitch usage rates into lower-rank factor matrices, NMF uncovers latent pitching styles that are strictly additive. The method represents each pitcher as a weighted combination of archetypes rather than assigning pitchers to a single category. This parts-based representation distinguishes NMF from techniques such as PCA and produces archetypes that correspond to intuitive, interpretable pitching philosophies.

The framework is computationally efficient, unsupervised, and requires no outcome labels which makes it particularly well-suited to data-scarce amateur environments. Beyond pitcher evaluation, NMF provides a generalizable approach to archetype discovery across sports analytics wherever player tendencies can be represented as non-negative matrices of usage rates, frequencies, or proportions.

1 Introduction

Pitcher evaluation in amateur baseball is fundamentally different from evaluation at the professional level. In summer collegiate leagues, seasons are extremely short, scouting resources are thin, and decisions often need to be made with incomplete information. Coaches and staff are forced to rely on subjective assessments, small sample performance stats, or informal comparisons that do not scale well across teams. From an analytics perspective, this creates a clear problem: we need methods that are fast and interpretable.

Rather than asking whether we can perfectly predict future performance, this paper asks a more foundational question: can we structure limited tracking data in a way that meaningfully compares pitchers and reveals underlying styles of play? The core research question is whether pitcher archetypes can be identified using sparse, imperfect pitch-tracking data, and whether those archetypes remain interpretable and useful in a real player development and scouting environment.

To answer this, we adapt Non-Negative Matrix Factorization (NMF), a dimensionality reduction technique with an established mathematical foundation, to the domain of amateur baseball pitching. The result is a lightweight, interpretable framework that requires only pitch-type usage data and produces archetypes that coaches and scouts can reason about directly, without statistical expertise.

The remainder of this paper is organized as follows. Section 2 reviews related work on pitcher classification at the professional level and identifies the gap this work addresses. Section 3 formally defines the problem setting and data representation. Section 4 presents the NMF methodology in full. Section 5 applies the framework to a case study of the 2025 California Collegiate League, and the later sections discusses the results, limitations, and directions for future work.

2 Background and Related Work

At the professional level, pitcher classification is well-studied. MLB teams frequently use Statcast data to cluster pitchers into archetypes such as high-spin fastball pitchers, sinker-heavy ground ball arms, or command-first profiles. Common methods include k-means clustering, Gaussian mixture models, and heuristic rules based on pitch characteristics. These approaches have proven effective at the MLB level, where seasons are long, sample sizes are large, and high-fidelity tracking data is available for every pitch thrown.

A key limitation of many clustering methods, however, is that they force pitchers into a single discrete category, ignoring that real pitchers often blend multiple styles. A pitcher may rely on a fastball-heavy approach when ahead in the count while switching to a breaking-ball-dominant strategy when behind — a nuance that hard cluster assignments cannot capture. Soft-assignment methods such as Gaussian mixture models partially address this, but their latent components are not constrained to be interpretable, and their outputs are not easily communicated to coaches or scouts in a practical setting.

There is relatively little work applying these classification techniques to amateur or developmental leagues. This is partly a data availability problem and partly a methodological one. Until recently, pitch-tracking systems like Trackman were limited to professional facilities. Amateur leagues have shorter seasons, smaller rosters, and far fewer pitches per pitcher, meaning that methods developed for MLB-scale data do not transfer cleanly. This creates a gap where interpretable, low-dimensional representations are especially needed. Decisions must be made quickly, with less data, by evaluators who may not have quantitative backgrounds.

In this work, we are not trying to invent a new model. Instead, we are trying to adapt an existing, mathematically rigorous approach to a setting where interpretability and practicality are the primary constraints. The methodology is directly inspired by Chapter 53 of *Mathletics* Winston et al. [2022] by Nestler, Pelechrinis, and Winston, in which the authors apply NMF to analyze NBA shooting patterns across different court areas. We port this framework to amateur baseball pitching, substituting court zones for count-state pitch-usage profiles and other pitcher attributes. Crucially, the resulting classifications must also pass the "eye test." They should align with how coaches and scouts already think about pitching styles, not just minimize a reconstruction error.

3 Problem Formulation

3.1 Setting

We consider a league consisting of n pitchers, each of whom has had their pitches tracked by a Trackman system over the course of a season. For each pitcher, we observe their pitch usage tendencies in different counts, pitch movement profiles, and pitch results. We define a *pitcher profile* as a vector of these usage rates, representing a compact summary of a pitcher's tendencies on the mound.

Formally, let there be p pitch types and c count states of interest. Each pitcher i is described by a profile vector $s_i \in \mathbb{R}_{\geq 0}^{p \cdot c}$, where entry (t, ℓ) records the rate at which pitcher i throws pitch type t in count state ℓ. Stacking these profiles as rows yields the data matrix:

$$S \in \mathbb{R}_{\geq 0}^{n \times m}, \qquad m = p \cdot c$$

All entries of S are non-negative by construction, as they represent empirical proportions bounded in $[0, 1]$.

3.2 Sparsity and Noise

In practice, S is sparse and noisy for several reasons. First, amateur seasons are short. A pitcher may accumulate only a few dozen tracked appearances, leading to high-variance estimates of their true usage rates, particularly in less common count states. Second, Trackman classification is imperfect: pitch types are algorithmically assigned and can be mislabeled, particularly for pitchers with non-standard grips or movement profiles.

These properties mean that any method applied to S must be robust to imperfect data and should not overfit to individual pitcher observations.

3.3 Goal

Given S, our goal is to identify a small number k of latent pitcher archetypes such that every pitcher in the dataset can be approximately described as a weighted combination of those archetypes. Formally, we seek matrices $W \in \mathbb{R}_{\geq 0}^{n\times k}$ and $H \in \mathbb{R}_{\geq 0}^{k\times m}$ such that:

$$S \approx W \cdot H$$

The rows of H define the archetypes themselves. Each row is a characteristic pitch-usage profile. The rows of W encode each pitcher's affinity for each archetype. We require both matrices to be non-negative so that the reconstruction is strictly additive: every pitcher is a blend of archetypes. This constraint is what makes the resulting decomposition interpretable in a baseball context, and it motivates the use of Non-Negative Matrix Factorization, which we describe in the following section.

4 Methodology: Non-Negative Matrix Factorization

We model pitcher profiling as a matrix approximation problem, seeking to recover a small number of latent archetypes that collectively explain the observed variation in pitching tendencies across a roster. The central tool is Non-Negative Matrix Factorization (NMF), a dimensionality reduction technique particularly well-suited to data that is non-negative by construction and where interpretability of the latent components is a first-class requirement.

4.1 Data Representation

We represent pitching data as a non-negative matrix $S \in \mathbb{R}_{\geq 0}^{n \times m}$, where each row corresponds to an individual pitcher and each column corresponds to a measurable pitching tendency, specifically we are looking at the usage rate of a given pitch type within a given count state. All entries are non-negative by construction, as they represent empirical usage rates bounded in $[0, 1]$.

4.2 Non-Negative Matrix Factorization

Non-Negative Matrix Factorization seeks to decompose S into two lower-rank non-negative matrices:

$$S_{n \times m} \approx W_{n \times k} \cdot H_{k \times m}$$

where $k \ll \min(n, m)$ is the number of latent components. This is achieved by solving the following constrained optimization problem:

$$\min_{W,H} \|S - WH\|_F^2 \quad \text{s.t.} \quad W_{i,j} \geq 0,\ H_{i,j} \geq 0 \quad \forall\, i, j \tag{1}$$

where $\| \cdot \|_F$ denotes the Frobenius norm:

$$\|A\|_F = \sqrt{\sum_{i,j} A_{ij}^2}$$

The optimization in Equation 1 is non-convex in W and H jointly, but is convex in each variable separately when the other is held fixed. In practice, this is solved via alternating multiplicative update rules Lee and Seung [1999], which iteratively refine W and H while preserving non-negativity at every step:

$$H \leftarrow H \odot \frac{W^\top S}{W^\top W H}, \qquad W \leftarrow W \odot \frac{S H^\top}{W H H^\top}$$

where $\odot$ denotes element-wise multiplication and all division is element-wise. These updates are guaranteed to be non-increasing in the objective and converge to a local minimum.

4.3 Interpretation of the Factor Matrices

The two factor matrices produced by NMF have distinct and complementary interpretations. $W \in \mathbb{R}_{\geq 0}^{n \times k}$ is the **weight matrix**, where each row W_i encodes how strongly pitcher i aligns with each of the k latent archetypes. $H \in \mathbb{R}_{\geq 0}^{k \times m}$ is the **loading matrix**, where each row H_j defines archetype j in terms of its characteristic pitch-type usage profile. Together, the product WH reconstructs the original data matrix as a weighted sum of these archetype profiles.

4.4 Why NMF and Not PCA?

A natural alternative for dimensionality reduction is Principal Component Analysis (PCA). However, PCA does not constrain its loadings or scores to be non-negative, meaning the resulting components can involve subtractions. In PCA, a pitcher's profile might be described as "high on component 1, minus component 3." These contrasts are mathematically valid but difficult to interpret in a scouting or player development context.

The non-negativity constraint in NMF enforces a *strictly additive* parts-based representation Lee and Seung [1999]: every pitcher is modeled as a non-negative weighted combination of archetypes, never a contrast between them. This means each latent component must correspond to a coherent, non-negative pitching profile that stands on its own. This makes the resulting archetypes directly interpretable to coaches and scouts without requiring statistical expertise.

4.5 Selecting the Number of Components

The rank k is a hyperparameter that must be specified prior to fitting. We select k by evaluating reconstruction error across a range of candidate values and applying the elbow method. Formally, let:

$$\mathcal{E}(k) = \|S - W^{(k)}H^{(k)}\|_F^2$$

As k increases, $\mathcal{E}(k)$ decreases monotonically, since a higher-rank factorization can always achieve at least as good a fit. However, beyond a certain point the marginal reduction in error becomes negligible while the interpretability of the components degrades as k grows large. We select the value of k at the inflection point of $\mathcal{E}(k)$, particularly the point at which additional components yield diminishing returns in reconstruction quality.

5 Case Study: 2025 California Collegiate League

We apply the NMF framework developed in Section 4 to pitching data from the 2025 California Collegiate League (CCL), a summer collegiate wood-bat league featuring players from NCAA programs across the country. Pitch-tracking data was collected via Trackman, providing pitch-type classifications and count-state information for each appearance.

5.1 Dataset Construction

We restrict the dataset to pitchers with a sufficient number of tracked pitches to produce stable usage rate estimates, yielding $n = 50$ pitchers. For each pitcher, we compute usage rates for three pitch types — fastball (FB), breaking ball (BB), and offspeed (CH) — in two count states: ahead and behind. This yields $m = 6$ features per pitcher, and we construct the data matrix $S \in \mathbb{R}_{\geq 0}^{50\times 6}$ as follows:

$$\underbrace{\begin{bmatrix} 0.393 & 0.446 & 0.161 & 0.803 & 0.136 & 0.061 \\ 0.460 & 0.287 & 0.253 & 0.871 & 0.057 & 0.071 \\ 0.333 & 0.424 & 0.242 & 0.761 & 0.183 & 0.056 \\ 0.357 & 0.333 & 0.310 & 0.685 & 0.278 & 0.037 \\ 0.363 & 0.375 & 0.263 & 0.511 & 0.311 & 0.178 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}}_{S_{50\times 6}}$$

where columns correspond to FB-Ahead, BB-Ahead, CH-Ahead, FB-Behind, BB-Behind, and CH-Behind usage rates respectively.

5.2 Selecting the Number of Archetypes

We fit NMF models for $k \in \{1, 2, \ldots, 10\}$ and evaluate reconstruction error $\mathcal{E}(k) = \|S - W^{(k)}H^{(k)}\|_F^2$ at each value, as shown in Figure 1.

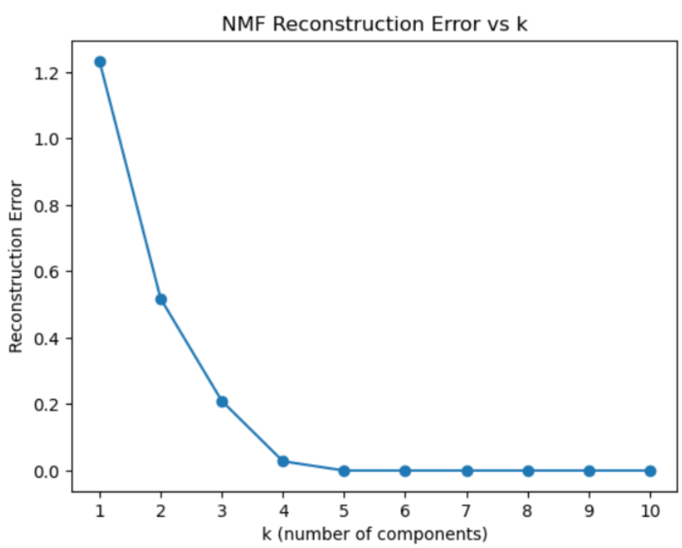

Figure 1: NMF reconstruction error as a function of the number of latent components k.

The curve drops steeply from $k = 1$ ($\mathcal{E} \approx 1.23$) to $k = 2$ ($\mathcal{E} \approx 0.52$) and again sharply to $k = 3$ ($\mathcal{E} \approx 0.21$), reflecting large gains in reconstruction quality as the first few archetypes are introduced. At $k = 4$, the error falls dramatically to near zero ($\mathcal{E} \approx 0.03$), and for all $k \geq 5$ the curve is essentially flat, with reconstruction error indistinguishable from zero at the scale of the plot. The transition from meaningful error reduction to negligible gain happens sharply and unambiguously at $k = 4$, making the choice of four archetypes well-supported by the data rather than a subjective judgment call. We therefore proceed with $k = 4$.

The sharp elbow at $k = 4$ warrants a brief discussion. Because pitch-type usage rates within each count state are compositional, they satisfy the constraints $\mathrm{FB}_{\mathrm{Ahead}} + \mathrm{BB}_{\mathrm{Ahead}} + \mathrm{CH}_{\mathrm{Ahead}} = 1$ and $\mathrm{FB}_{\mathrm{Behind}} + \mathrm{BB}_{\mathrm{Behind}} + \mathrm{CH}_{\mathrm{Behind}} = 1$. As a result, the six columns of S are not linearly independent. These two constraints reduce the intrinsic dimensionality of the data from six to four, implying that a rank-4 factorization is theoretically sufficient to reconstruct the matrix with minimal error. From this perspective, the pronounced elbow at $k = 4$ is not unexpected. Rather than reflecting an arbitrary modeling choice, it arises directly from the structure of the underlying data representation. The result also provides a useful sanity check: despite receiving no information about the true dimensionality of the problem, the NMF procedure correctly identifies the intrinsic rank implied by the compositional nature of the data.

5.3 The Factorization

Applying NMF with $k = 4$ to S yields the following decomposition $S \approx W \cdot H$:

$$\underbrace{\begin{bmatrix} 1.326 & 0.306 & 0.186 & 0.047 \\ 1.143 & 0.024 & 0.304 & 0.155 \\ 0.966 & 0.358 & 0.293 & 0.081 \\ 0.351 & 0.394 & 0.377 & 0.257 \\ 0.000 & 0.529 & 0.337 & 0.326 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}}_{W_{50\times 4}} \times \underbrace{\begin{bmatrix} 0.254 & 0.173 & 0.009 & 0.452 & 0.000 & 0.000 \\ 0.000 & 0.520 & 0.000 & 0.073 & 0.434 & 0.000 \\ 0.000 & 0.255 & 0.793 & 0.828 & 0.000 & 0.203 \\ 1.089 & 0.000 & 0.000 & 0.650 & 0.294 & 0.105 \end{bmatrix}}_{H_{4\times 6}}$$

The rows of H define the four archetypes in terms of their characteristic pitch-usage profiles, and the rows of W encode each pitcher's affinity for each archetype. We interpret each archetype in turn below.

5.4 Archetype Interpretation

Each row of H corresponds to one latent archetype. We interpret each by examining which features (pitch types in which count states) carry the largest loadings.

Table 1: Archetype loading matrix $H_{4\times6}$. Largest loading per row is bolded.

Archetype	FB-Ah.	BB-Ah.	CH-Ah.	FB-Bh.	BB-Bh.	CH-Bh.
1. Ahead Diversification / Heater Recovery	0.254	0.173	0.009	**0.452**	0.000	0.000
2. Breaking-Ball Dominant	0.000	**0.520**	0.000	0.073	0.434	0.000
3. Offspeed-Ahead / Fastball-Behind	0.000	0.255	**0.793**	0.828	0.000	0.203
4. Fastball Establisher / Behind Diversification	**1.089**	0.000	0.000	0.650	0.294	0.105

Archetype 1 — Ahead Diversification / Heater Recovery. This archetype is characterized by a high fastball usage rate when behind in the count (loading: 0.452) and moderate diversification when ahead. Pitchers who load heavily onto this component tend to attack hitters with fastballs when they need a strike, reverting to a heater-first approach under pressure, while mixing more freely when in control of the at-bat. At the MLB level, this profile is reminiscent of pitchers like **Zack Wheeler**, who consistently elevates his four-seam fastball usage in two-strike and behind counts. In our dataset, Pitcher 1 (row 1 of W, weight: 1.326) is the strongest exemplar of this archetype.

Archetype 2 — Breaking-Ball Dominant. This archetype loads heavily on breaking ball usage both ahead (0.520) and behind (0.434) in the count, with near-zero fastball and offspeed loadings. Pitchers in this group use their breaking ball as a primary weapon regardless of count state which is a relatively uncommon profile at the amateur level, where fastballs typically dominate. This profile is analogous to MLB pitchers like **Clayton Kershaw** in his prime, whose curveball was a go-to offering in nearly every count. Pitcher 5 (row 5 of W, weight: 0.529) loads most strongly on this archetype and not at all on Archetype 1, suggesting a pitcher who has fully committed to a breaking-ball-first identity.

Archetype 3 — Offspeed-Ahead / Fastball-Behind. This archetype is defined by heavy offspeed usage when ahead in the count (CH-Ahead loading: 0.793) and a pivot to fastballs when behind (FB-Behind loading: 0.828). Pitchers in this group use offspeed pitches to put hitters away when ahead, but revert to fastballs to regain the strike zone when behind. A comparable MLB profile is **Kyle Hendricks**, who consistently tunnels his changeup in favorable counts and relies on fastball location when he needs a strike. Pitcher 4 (row 4 of W) shows meaningful weight on both Archetypes 2 and 3 (0.394 and 0.377 respectively), illustrating the blending property of NMF and that pitchers draw from multiple styles.

Archetype 4 — Fastball Establisher / Behind Diversification. This archetype has the largest single loading in the entire H matrix: a fastball-ahead usage of 1.089. Pitchers here establish the fastball aggressively when ahead, then introduce breaking balls and offspeed when behind (BB-Behind: 0.294, CH-Behind: 0.105). This is perhaps the most classically "power pitcher" archetype in the dataset. At the MLB level, this resembles **Gerrit Cole**, who leads with his four-seam fastball to set up secondary offerings later in counts.

5.5 The Additive Blending Property in Practice

A key advantage of NMF over hard-clustering methods is visible directly in the W matrix. Rather than assigning each pitcher to exactly one archetype, NMF produces a continuous weight vector for each pitcher across all four components. Pitcher 4, for example, carries weights of 0.351, 0.394, 0.377, and 0.257 across all four archetypes, an even blend suggesting a highly adaptable pitcher with no single dominant style. By contrast, Pitcher 1 loads almost entirely on Archetype 1 (weight: 1.326) with near-zero weights elsewhere, indicating a pitcher whose tendencies are almost entirely explained by the Heater Recovery profile. This spectrum is information that hard clustering simply cannot provide, and it is precisely the kind of nuance that is useful in a player development context.

6 Results and Validation

6.1 Reconstruction Quality

The NMF decomposition with $k = 4$ achieves a reconstruction error of $\mathcal{E}(4) \approx 0.03$, representing a 97.6% reduction in error relative to the single-component baseline ($\mathcal{E}(1) \approx 1.23$). As discussed in Section 5.2, the error curve drops sharply through $k = 3$ and then falls to near zero at $k = 4$, after which additional components provide no meaningful improvement. This suggests that four archetypes are not only sufficient but nearly complete in their ability to reconstruct the observed pitching tendencies across the roster. The data itself is telling us that four latent styles underlie the variation we observe.

6.2 Interpretability as a First-Class Goal

A model that minimizes reconstruction error but produces uninterpretable components is of limited value in a scouting or player development context. We therefore evaluate the four archetypes not only on statistical grounds but on whether they pass the "eye test", whether they correspond to pitching styles that coaches and scouts would recognize and find useful.

Each of the four archetypes identified is grounded in a coherent, recognizable pitching philosophy. Archetype 1 captures the classic "fastball under pressure" tendency, a well-documented behavioral pattern in which pitchers revert to their most trusted pitch when behind in the count. Archetype 2 identifies a breaking-ball-first approach that is immediately recognizable to any pitching coach as a distinct and deliberate style. Archetype 3 reflects a sophisticated sequencing strategy, using offspeed pitches to put hitters away when ahead and fastballs to recover the strike zone when behind, that experienced scouts would associate with command-oriented pitchers. Archetype 4 describes the classic power pitcher profile: establish the fastball early, diversify later.

Crucially, none of these archetypes needed to be defined in advance. They emerged directly from the data via the NMF factorization, without any labels, rules, or human supervision. The fact that they are immediately legible to a baseball audience is a validation of both the method and the data representation.

6.3 The Blending Property as a Scouting Tool

Beyond the archetypes themselves, the weight matrix W provides a continuous pitcher-level profile that hard clustering methods cannot. Rather than assigning each pitcher to a single category, NMF quantifies the extent to which a pitcher exhibits each archetype. This allows analysts, coaches, and scouts to move beyond discrete labels and evaluate pitching style as a spectrum.

To make this concrete, consider a pitcher who loads almost entirely onto Archetype 1, Heater Recovery. Such a pitcher relies heavily on his fastball when behind in the count, making his pitch selection increasingly predictable under pressure. An opposing analytics staff could use this information to recommend that hitters sit on the fastball in 1-0 or 2-1 counts. From a player development perspective, the same profile suggests a clear objective: develop a secondary offering that can be trusted in hitter-friendly counts, shifting the pitcher toward a more balanced archetype blend such as Archetype 4, where breaking balls and offspeed pitches remain part of the repertoire even when behind.

At the opposite end of the spectrum is a pitcher with relatively uniform weights across all four archetypes, such as Pitcher 4 in our dataset. This pitcher is more difficult to exploit because no single tendency dominates his decision making. Rather than defaulting to a comfort zone under pressure, he appears capable of varying pitch selection across count states. For scouts, this balanced archetype profile is informative in its own right, suggesting a pitcher who sequences deliberately and may possess a more adaptable approach as competition levels increase.

Importantly, translating these results into baseball decisions requires little statistical expertise. The archetypes provide an intuitive vocabulary for describing pitching behavior, while the weight matrix quantifies how strongly each behavior is expressed. Together, they can inform matchup preparation, identify exploitable tendencies, establish developmental goals, and track how a pitcher's strategic profile evolves over time.

7 Discussion and Limitations

7.1 Generalizability of the Method

One of the most important results of this work is not the four specific archetypes identified in the 2025 CCL, but the demonstration that NMF is a natural and effective tool for this class of problem in sports analytics. The framework requires only that the data matrix S be non-negative, which is a condition that is satisfied by an extremely wide range of sports data types. Pitch-type usage rates are one instantiation, but the same approach applies directly to:

- **Batted ball profiles** — exit velocity and launch angle distributions across pitch types or zones, yielding hitter archetypes rather than pitcher archetypes.
- **Zone usage rates** — pitch location frequency across regions of the strike zone, providing a spatial complement to the count-state representation used here.
- **Defensive positioning data** — shift tendencies or fielder positioning frequencies across batter types, enabling archetypes for defensive schemes.
- **Workload and usage patterns** — pitch mix variation across game situations, innings, or opponent handedness.

In each case, the analyst constructs a non-negative matrix where rows are subjects and columns are features, applies NMF, and reads off interpretable archetypes from the rows of H. The mathematical machinery is identical; only the domain changes. This generality is what makes NMF a genuinely useful addition to the sports analytics toolkit, rather than a one-off application to a specific dataset.

7.2 Limitations

We acknowledge several limitations of the current work.

Small sample sizes. Amateur leagues have short seasons, and many pitchers in our dataset accumulate relatively few tracked pitches. Usage rate estimates computed from small samples are high-variance, and the entries of S may not accurately reflect a pitcher's true tendencies. This is an inherent constraint of the amateur setting, not a flaw in the method, but it means that archetype assignments for pitchers with limited data should be interpreted with appropriate caution.

Single-season data. Our analysis is based on a single season of the 2025 CCL. We cannot assess whether the archetypes identified are stable across seasons or whether individual pitchers' archetype loadings change as they develop. Longitudinal analysis across multiple seasons would substantially strengthen the validity of the framework.

No outcome variables. The NMF factorization is entirely unsupervised. It identifies structure in pitching tendencies without any reference to outcomes such as ERA, FIP, strikeout rate, or whiff rate. As a result, we cannot claim that any archetype is more or less effective than another. The framework tells us *how* pitchers pitch, not *how well*. Incorporating outcome variables as a validation layer is an important direction for future work. For example, we could test whether archetype membership predicts performance differences.

Feature design. The choice to represent pitchers by pitch-type usage in ahead and behind counts is one of many possible feature representations. Different column choices, finer count state granularity, the inclusion of pitch movement or velocity, or zone-based features, would yield different archetypes. The archetypes are a function of both the method and the representation, and future work should explore the sensitivity of results to feature design choices.

8 Conclusion

This paper has demonstrated that Non-Negative Matrix Factorization provides a principled, interpretable, and practically useful framework for identifying pitcher archetypes in amateur baseball. Applied to pitch-tracking data from the 2025 California Collegiate League, the method recovers four latent archetypes — Ahead Diversification / Heater Recovery, Breaking-Ball Dominant, Offspeed-Ahead / Fastball-Behind, and Fastball Establisher / Behind Diversification — each of which is mathematically grounded, emerges without supervision, and is immediately legible to a baseball audience. The reconstruction error falls to near zero at $k = 4$ and remains flat thereafter, providing strong quantitative support for the choice of four components.

The broader contribution of this work is methodological. We have shown that a classical matrix factorization technique, adapted from an application in NBA shooting analysis Winston et al. [2022], transfers cleanly to a new sport, a new level of play, and a new data representation. The success of this transfer reflects the fact that NMF is well-matched to the structural properties of sports data more generally. Wherever an analyst can construct a non-negative matrix of usage rates, frequencies, or proportions, NMF offers a ready-made framework for recovering interpretable latent structure. The archetypes identified in this paper are one instance of a much larger class of analyses that this approach makes possible.

For amateur baseball specifically, the implications are practical and immediate. In an environment where scouting resources are thin, seasons are short, and decisions must be made quickly from imperfect data, a method that is fast, unsupervised, and produces human-readable output is exactly what is needed. Coaches do not need a statistics degree to use the output of this model; the archetypes speak in the language of pitching.

Future work should extend this framework in several directions. Longitudinal analysis across multiple seasons would allow us to track how pitchers' archetype profiles evolve over time, potentially flagging developmental progress or regression. Incorporating outcome variables such as strikeout rate or contact quality would allow the archetypes to be evaluated not just descriptively but predictively. Expanding the feature representation to include pitch movement, velocity bands, or zone tendencies would enrich the profiles and potentially reveal finer-grained archetypes. And applying the same framework to hitting, fielding, or bullpen usage data would demonstrate the full generality of the approach across the sport.

At its core, this project is a case study in what applied mathematics can offer sports analytics when the goal is not prediction but understanding. NMF does not tell us who will be a good pitcher. It tells us, clearly and interpretably, what kind of pitcher someone already is, and in a development environment, that is often the more valuable question.

References

Daniel D. Lee and H. Sebastian Seung. Learning the parts of objects by non-negative matrix factorization. *Nature*, 401:788–791, 1999.

Wayne L. Winston, Scott Nestler, and Konstantinos Pelechrinis. *Mathletics: How Gamblers, Managers, and Fans Use Mathematics in Baseball, Basketball, and Football.* Princeton University Press, 2nd edition, 2022.